



JAIN (Deemed-to-be University)

Department of Computer Science and Engineering

**DATASET COLLECTION AND LITERATURE
REVIEW**

**PRODUCT PRICE PREDICTION AND COMPARISON
USING MACHINE LEARNING**

Selected Problem: Product Price Prediction and
Comparison Using Machine Learning

Dataset: Comparing Product price on various Online platform

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Abstract

E-commerce has created a highly competitive pricing environment in which the same or comparable products can be listed at different prices across online platforms. Historical price observations provide an opportunity to study price variation, platform differences and temporal patterns using machine learning. This literature review focuses on an ML-heavy product price prediction problem in which historical product prices, platform identity and date-related information are treated as the principal analytical signals.

The proposed project uses the provided *product prices.xlsx* dataset. After restructuring the spreadsheet, the dataset contains 468 product-date observations, 12 products, 10 online platforms and 3,237 observed platform-price records. The observations span 16 September 2024 to 24 October 2024. Several platforms have incomplete coverage, so missing values and availability must be handled explicitly before model development.

Twenty research papers and reviews were selected around e-commerce price prediction, price comparison, dynamic pricing, machine learning regression, time-series forecasting and reinforcement learning. The literature includes linear and tree-based regression, ensemble boosting, LSTM/deep learning, multimodal price estimation and reinforcement-learning approaches. Recent studies show continued interest in Random Forest, gradient boosting, XGBoost, time-series models and deep learning for pricing-related prediction, while dynamic pricing research increasingly considers competitor prices, demand and market state.

The proposed implementation will compare baseline and ensemble regression models, evaluate them using MAE, RMSE, R^2 and optionally RMSLE, and use time-aware validation to reduce temporal leakage. The project will also include platform-wise price comparison and feature-importance analysis. Deep learning and reinforcement learning are treated as future extensions because the current dataset is relatively small and does not contain demand, inventory or customer-behaviour variables.

Keywords: E-Commerce, Product Price Prediction, Price Comparison, Machine Learning, Regression, Random Forest, XGBoost, Dynamic Pricing, Time Series, Online Platforms.

1. Introduction

Online retail platforms continuously expose consumers to multiple sellers, delivery services and marketplace options. For the same product, the observed price can vary across platforms and across dates. This makes price comparison a useful data-analytics problem and creates a basis for forecasting future prices from historical observations.

1.1 E-Commerce Pricing

E-commerce pricing is influenced by product characteristics, platform strategy, discounts, competition, demand, inventory and temporal effects. Research on online pricing treats pricing as both a prediction problem and a decision problem. Product-price prediction estimates a numerical price, while dynamic pricing determines how a seller or platform should change price in response to market conditions.

1.2 Price Comparison

A price-comparison system collects or receives prices from multiple platforms and presents differences in a common structure. Research on e-commerce comparison portals demonstrates the usefulness of extracting product information, cleaning noisy values and comparing offers across platforms. Such systems can be extended with predictive analytics to estimate future price trends.

1.3 Machine Learning for Price Prediction

Price prediction is naturally formulated as a supervised regression problem when the target variable is numerical. Linear Regression provides a simple baseline, Decision Trees capture nonlinear rules, Random Forest aggregates many trees, and boosting methods such as XGBoost learn complex nonlinear relationships. Deep learning can model sequential or multimodal information when sufficient data are available.

1.4 Project Focus

The selected project is intentionally focused on machine learning and data analysis. The first stage uses the supplied historical dataset rather than real-time web scraping. The main technical tasks are data cleaning, restructuring, exploratory analysis, feature engineering, time-aware validation, regression-model comparison and interpretation. Price comparison is the analytical layer, while prediction is the ML layer.

2. Problem Statement, Aim and Objectives

2.1 Problem Statement

Online platforms can show different prices for the same product, and those prices can change over time. Manually comparing historical prices does not provide a systematic way to estimate future prices. The problem is to develop a machine-learning framework that learns from historical product, platform and time information to predict product price while supporting cross-platform comparison.

2.2 Aim

To develop and evaluate a machine-learning-based framework for predicting product prices across online platforms and analyzing historical price differences using a multi-platform e-commerce price dataset.

2.3 Objectives

- Collect and document the product-price dataset and its platform structure.
- Review research on e-commerce price prediction, price comparison and dynamic pricing.
- Clean and transform the spreadsheet into an analysis-ready long-format dataset.
- Engineer product, platform and temporal features and investigate lag-based features.
- Compare baseline and ensemble regression models for price prediction.
- Evaluate models using MAE, RMSE and R^2 , with RMSLE where appropriate.
- Analyze price variation and platform-level differences.
- Identify research gaps and define future extensions such as real-time data collection and sequential deep learning.

2.4 Research Questions

1. Which machine-learning methods are commonly used for e-commerce product price prediction?
2. How can historical prices across multiple online platforms be transformed into predictive features?
3. Which regression models are suitable for the structured dataset used in this project?
4. How should missing platform prices and temporal leakage be handled?
5. What research gaps remain for real-time multi-platform price prediction?

3. Dataset Collection

3.1 Product Price Dataset

The selected dataset is the Kaggle dataset titled *Comparing Product price on various Online platform*. The local spreadsheet analyzed for this report is *product prices.xlsx*. The original worksheet uses a two-level header in which the platform names are listed across the price columns. For analysis, the headers were normalized to Date, SIno, Product, Jio Mart, Blinkit, BigBasket, Zepto, Swiggy, Licious, Fresh To Home, Country Delight, DMart and Amazon Fresh.

3.2 Dataset Profile

Property	Value
Rows after cleaning header rows	468 product-date observations
Unique products	12
Online platforms	10
Observed platform-price records	3,237
Date range	16 Sep 2024 – 24 Oct 2024
Unique dates	39
Original file	product prices.xlsx
Primary target for ML	Price
Problem type	Supervised regression

3.3 Products

1	POTATO (1kg)
2	TOMATO (1kg)
3	BANANA (1kg)
4	PARLE-G GLUCOSE BISCUITS (800g)
5	SUNDROP LITE SUNFLOWER REFINED OIL (1L)
6	SURF EXCEL EASY WASH DETERGENT POWDER (1kg)
7	AMUL PURE GHEE (1L)
8	AASHIRVAAD HIGH FIBRE ATTA WITH MULTIGRAINS (5kg)
9	AMUL TAAZA HOMOGENISED TONED MILK TETRA PACK (1L)
10	AMUL BUTTER CARTON (500g)
11	CHICKEN (1kg)
12	EGGS (1 DOZEN)

3.4 Platform Coverage

Platform	Observed prices	Coverage of 468 rows	Mean observed price (₹)
Jio Mart	429	91.7%	180.08
Blinkit	429	91.7%	186.53
BigBasket	468	100.0%	189.94

Platform	Observed prices	Coverage of 468 rows	Mean observed price (■)
Zepto	390	83.3%	183.93
Swiggy	468	100.0%	200.38
Licious	78	16.7%	202.60
Fresh To Home	78	16.7%	222.99
Country Delight	117	25.0%	62.51
DMart	351	75.0%	198.07
Amazon Fresh	429	91.7%	196.60

The platform means are descriptive only. They should not be interpreted as a general ranking of platforms because the platforms have different product coverage and different missingness patterns. In particular, Licious, Fresh To Home and Country Delight have substantially fewer observations in this dataset.

3.5 Main Variables

Variable	Role	Meaning
Date	Feature	Observation date; used for temporal features
Product	Feature	Product identity/category proxy
Platform	Feature	Online platform identity
Price	Target	Observed product price in ■
Lagged Price	Derived Feature	Previous observed price for product/platform
Price Statistics	Derived Features	Rolling mean, minimum, maximum or change where valid

3.6 Dataset Limitation

The dataset is relatively small and covers only 39 dates. Platform availability is uneven, and the source spreadsheet uses missing markers such as “-” as well as blank cells. The data therefore support a student-level prototype and comparative experiment, but results should not be treated as a universal representation of online retail pricing. The short time window also limits the ability to learn long seasonal cycles.

4. Dataset Preparation and ML Pipeline

4.1 Data Cleaning

- Remove the two-level header structure and retain one normalized header row.
- Forward-fill dates because the source worksheet records the date once for each 12-product block.
- Remove non-data/header rows.
- Convert “-” and blank entries to missing values.
- Convert price fields to numeric values.
- Reshape the platform columns into a long-format table with Date, Product, Platform and Price.
- Check duplicates, impossible values and inconsistent product labels.

4.2 Exploratory Data Analysis

EDA should examine product-level price distributions, platform-level distributions, daily price movement, missingness, outliers and cross-platform differences. Histograms, box plots, line plots and correlation matrices can reveal the structure of the data. Because products have very different natural price levels, comparisons should also be made within product rather than only across the overall price column.

4.3 Feature Engineering

- Product encoding using one-hot encoding or target-safe categorical encoding.
- Platform encoding using one-hot encoding.
- Calendar features: day of month, day of week and elapsed days.
- Lag-1 and lag-k prices for each product-platform series when enough observations exist.
- Price change and percentage change from the previous observation.
- Rolling mean/median features computed only from past observations to avoid leakage.
- Cross-platform features when the prediction setting assumes competitor prices are known at prediction time.

4.4 ML Workflow

1. Load and normalize the spreadsheet.
2. Clean missing values and convert to long format.
3. Perform EDA and missingness analysis.
4. Create time, product and platform features.
5. Construct lagged features using past data only.
6. Use a chronological train-validation-test split.
7. Train baseline regression models.
8. Train Random Forest and XGBoost/gradient boosting models.
9. Compare MAE, RMSE and R^2 ; optionally RMSLE.
10. Inspect residuals and feature importance.
11. Document limitations and define future real-time extensions.

5. Literature Review Methodology

The literature search was organized around e-commerce product price prediction, online price comparison, dynamic pricing, machine-learning regression, time-series forecasting and reinforcement learning. Foundational work was retained where it established an important pricing formulation, while recent studies were included to capture current machine-learning and deep-learning directions.

5.1 Inclusion Criteria

- Direct relevance to e-commerce price prediction, price comparison or dynamic pricing.
- Use of machine learning, artificial intelligence, regression, forecasting or reinforcement learning.
- Peer-reviewed journal, established conference or identifiable scholarly publication.
- Studies reporting methods, datasets, evaluation metrics or deployment considerations relevant to the project.
- Recent literature from 2023–2026 included to reflect current research directions, with selected earlier foundational studies.

5.2 Review Themes

- Product-price regression.
- Cross-platform price comparison.
- Dynamic pricing and competitor effects.
- Tree ensembles and boosting.
- Time-series and deep-learning forecasting.
- Reinforcement learning for sequential pricing decisions.
- Multimodal product-price estimation.
- Data sparsity, cold-start and generalization.

5.3 Evidence Base

Twenty selected studies were used for the review. The evidence includes ACM, IEEE, Elsevier/ScienceDirect, Springer, MDPI and other scholarly venues. The studies range from direct product-price regression to broader dynamic-pricing research. The review distinguishes prediction of an observed price from optimization of a future price decision.

6. Machine Learning Methods for Product Price Prediction

6.1 Linear Regression

Linear Regression models price as a weighted combination of input variables. It is useful as a transparent baseline and can indicate whether simple additive relationships explain a meaningful portion of price variation. Its main limitation is difficulty representing complex nonlinear interactions.

6.2 Decision Tree Regressor

A Decision Tree Regressor recursively splits observations using feature thresholds and predicts a value at each leaf. It can model nonlinear relationships and interactions but may overfit if depth and leaf-size constraints are not controlled.

6.3 Random Forest Regressor

Random Forest combines many decision trees trained with randomized samples and feature subsets. It is suitable for structured tabular data and can model nonlinear interactions. Several e-commerce price-prediction studies report strong results from Random Forest models, although performance is dataset-dependent.

6.4 Gradient Boosting and XGBoost

Boosting methods build an ensemble sequentially, where later trees focus on errors from earlier learners. XGBoost is widely used for structured regression because it can capture nonlinear interactions and supports regularization and tuning. It is a strong candidate for the proposed project.

6.5 Support Vector Regression

Support Vector Regression can model nonlinear relationships using kernel functions. It can be effective for smaller datasets but generally requires feature scaling and careful selection of hyperparameters.

6.6 Model Selection

The proposed comparison should begin with Linear Regression and Decision Tree as interpretable baselines, followed by Random Forest and XGBoost or another gradient-boosting model. SVR may be included as an additional comparison. The final model should be selected from measured validation results rather than assumed in advance.

7. Deep Learning, Time Series and Advanced Pricing

7.1 LSTM

Long Short-Term Memory networks are recurrent neural networks designed to learn dependencies across sequences. They become relevant when a price series has sufficient length and when recent historical observations provide predictive information.

7.2 Attention and BiLSTM

Attention mechanisms allow models to assign different importance to parts of an input sequence. Recent e-commerce dynamic-pricing work combines bidirectional LSTM and attention for time-dependent pricing signals. Such approaches are technically relevant but require more data and careful temporal validation than the present dataset provides.

7.3 Reinforcement Learning

Reinforcement learning formulates pricing as a sequential decision problem in which an agent selects actions and receives rewards. Research has applied DQN, SAC, PPO and other approaches to dynamic pricing. This is conceptually different from the current project, which predicts observed price rather than directly optimizing a pricing action.

7.4 Multimodal Price Estimation

Recent work also combines product images, textual descriptions and categorical attributes for price regression. These approaches are useful when the dataset contains rich product metadata, but the current spreadsheet contains primarily product, platform, date and price information. Therefore, multimodal models are a future extension rather than an immediate requirement.

7.5 Relevance to This Project

The current dataset supports a structured regression problem. Tree-based models are therefore a practical starting point. LSTM, BiLSTM, attention and reinforcement learning should be treated as future extensions after collecting a larger, richer and more continuous dataset.

8. Literature Review: Papers 1–5

1. Ferreira, Lee & Simchi-Levi (2018) – Powering retailers’ digitization through analytics and automation

The study discusses analytics and automation across retail activities, including price optimization, online retail and forecasting. It establishes the broader role of analytics in retail decision-making and supports the project’s emphasis on data-driven pricing.

2. Wang et al. (2024/2025) – Machine Learning Study on Cross-border E-commerce Products Price Prediction: Evidence from Platform of Lazada

This study formulates e-commerce product pricing as supervised regression and compares ordinary least squares, Random Forest and XGBoost. The authors report Random Forest as the strongest among the tested methods and use feature analysis to understand pricing factors. It is directly relevant to the proposed product-price regression task.

3. Namburu, Selvaraj & Varsha (2022) – Product pricing solutions using hybrid machine learning algorithm

The paper proposes X-NGBoost, combining XGBoost and NGBoost, for product-price prediction using product, review, statistical and categorical information. It compares the hybrid approach with boosting ensembles and demonstrates the potential of advanced ensemble regression for e-commerce pricing.

4. Kaur (2023) – Price Prediction in E-Commerce Using Machine Learning Models: A Comparative Study

The study compares Linear Regression, Decision Tree Regressor, Random Forest Regressor and MLP Regressor for e-commerce final price prediction. It uses R^2 and RMSE and reports Random Forest as the strongest among the evaluated models. This supports including Random Forest as a primary baseline/ensemble candidate.

5. Sharma et al. (2022) – E-Commerce product comparison portal for classification of customer data based on data mining

The work extracts product names, prices, reviews and ratings from multiple e-commerce platforms and cleans the resulting data for comparison. It demonstrates the practical value of multi-platform product data and supports the price-comparison component of the proposed project.

9. Literature Review: Papers 6–10

6. Amirtharaj & Shiva Varshini (2025) – Evaluation of Machine Learning Based Prediction Systems for Comparing Online Product Prices

This work combines multi-platform price extraction and ML forecasting, comparing Linear Regression, Decision Tree, Random Forest and XGBoost. Its problem formulation is closely related to the present project because it links price comparison with future price prediction.

7. Esclanda-Lo, Ong & Masangkay (2024) – E-Commerce price index prediction with time series mining and automated machine learning

The study develops a time-sensitive pipeline using e-commerce price data to construct an alternative price index. It demonstrates that e-commerce price observations can support time-series analysis and broader price measurement, reinforcing the importance of temporal structure.

8. Guo & Zhang (2025) – Dynamic Pricing Models in E-Commerce: Exploring Machine Learning Techniques to Balance Profitability and Customer Satisfaction

The study reviews machine-learning approaches to dynamic pricing and presents an LSTM-based framework using historical prices, demand and external factors such as competitor pricing. It highlights the distinction between forecasting prices and optimizing pricing decisions.

9. Nowak & Pawłowska-Nowak (2024) – Dynamic Pricing Method in the E-Commerce Industry Using Machine Learning

This work presents a machine-learning approach to dynamic pricing and illustrates the use of data-driven methods for adaptive pricing. It is relevant to future extensions where the project moves from predicting observed prices toward pricing recommendations.

10. Dritsas & Trigka (2025) – Machine Learning in E-Commerce: Trends, Applications, and Future Challenges

This IEEE Access survey covers machine learning across e-commerce, including dynamic pricing, recommendation, fraud detection and customer analytics. It identifies scalability, interpretability, data sparsity and related issues that are also important when extending the present prototype.

10. Literature Review: Papers 11–15

11. Liu et al. (2019) – Dynamic Pricing on E-commerce Platform with Deep Reinforcement Learning: A Field Experiment

The authors formulate dynamic pricing as a Markov Decision Process and evaluate deep reinforcement learning using real e-commerce data and field experiments. The study is important because it shows how pricing can be treated as a sequential decision problem rather than only as regression.

12. Yin & Han (2021) – Dynamic Pricing Model of E-Commerce Platforms Based on Deep Reinforcement Learning

The paper develops a dynamic pricing framework using deep reinforcement learning and game-theoretic analysis for e-commerce platforms. It provides a theoretical foundation for competitive and adaptive pricing but is beyond the immediate scope of the current regression task.

13. Kastius & Schlosser (2022) – Dynamic Pricing Under Competition Using Reinforcement Learning

This research evaluates reinforcement-learning approaches such as DQN and SAC under competitive market settings. It emphasizes that competitor interaction changes the pricing problem and motivates future work using cross-platform information.

14. Liu et al. (2023) – Enhancing Sparse Data Performance in E-Commerce Dynamic Pricing with Reinforcement Learning and Pre-Trained Learning

The study addresses sparse-data and cold-start problems in e-commerce dynamic pricing and proposes product clustering and pre-trained learning ideas. Its discussion of sparse observations is relevant because the present dataset has uneven platform coverage.

15. Mussi et al. (2023) – Dynamic Pricing with Volume Discounts in Online Settings

The study considers online pricing with transaction data and dynamic decisions under volume discounts. It demonstrates the broader class of online learning and pricing problems in which the objective extends beyond predicting a static price.

11. Literature Review: Papers 16–20

16. Zhang, Zheng & Qi (2018) – Deep Learning for E-Commerce Price Prediction

This ACM KDD study investigates deep learning for e-commerce price prediction and demonstrates the use of neural representations for a pricing regression task. It provides an early deep-learning reference for the project and supports treating neural sequence or representation-learning methods as a possible extension when richer and larger data become available.

17. Redwan et al. (2025) – A Multimodal Deep Learning Framework for Integrating Visual, Textual and Categorical Features in Retail Price Estimation

The study combines image, text and categorical information using deep-learning architectures for retail price regression. It reports MAE, RMSE and R^2 and shows how multimodal information can improve price estimation. The approach suggests a future direction if product descriptions or images are added.

18. Padmashini, Rajesh Kumar & Ravi (2026) – A Machine Learning Approach for E-Commerce Sales Forecasting and Price Optimization in the Digital Society Era

The study combines XGBoost, LightGBM, Random Forest and MLP for e-commerce sales forecasting and pricing-related analysis. It reinforces the value of ensemble learning for nonlinear e-commerce data and highlights the importance of feature importance and evaluation metrics.

19. Liu et al. (2026) – A Multimodal Deep Reinforcement Learning Framework for Dynamic Pricing Optimisation in E-Commerce

This recent IEEE Access work integrates product multimodal context into a deep reinforcement-learning pricing framework. It illustrates the movement toward models that combine heterogeneous product information with sequential pricing decisions and provides a possible long-term extension path.

20. Ozay, Jahanbakht & Wang (2026) – AI-Based Dynamic Pricing: A Cross Industry Bibliometric Review of Trends, Challenges, and Future Directions

This recent review synthesizes AI-based dynamic-pricing research and identifies demand learning, pricing optimization and strategic interaction as major themes. It provides current context for the project and reinforces the distinction between predictive price estimation and full dynamic-pricing optimization.

12. Comparative Analysis of the 20 Studies

Paper	Primary Method	Data / Context	Main Task	Project Relevance
1	Retail analytics	Online retail	Analytics / price optimization	Analytics foundation
2	OLS / RF / XGBoost	Lazada e-commerce	Price prediction	Direct regression comparison
3	X-NGBoost	E-commerce product data	Price prediction	Hybrid ensemble
4	LR / DT / RF / MLP	E-commerce	Price regression	Baseline/model comparison
5	RPA + data mining	Amazon / Flipkart	Product comparison	Multi-platform cleaning
6	LR / DT / RF / XGBoost	Amazon / Flipkart	Price forecasting	Closest price-comparison formulation
7	Time-series mining + AutoML	E-commerce price data	Price index prediction	Temporal analysis
8	LSTM / ML	Dynamic e-commerce	Dynamic pricing	Future sequence model
9	ML dynamic pricing	E-commerce	Pricing method	Future pricing extension
10	ML survey	E-commerce	Review	Research context
11	Deep RL	Alibaba/Tmall	Dynamic pricing	Future decision model
12	Deep RL + game theory	E-commerce	Dynamic pricing	Competitive pricing context
13	DQN / SAC	Competitive markets	Dynamic pricing	Competitor effects
14	RL + pre-training	Sparse e-commerce	Dynamic pricing	Missing/sparse data issue
15	Online learning	Online settings	Dynamic pricing	Sequential decision context
16	OLS / RF / XGBoost	Lazada	Price prediction	Feature attribution
17	Multimodal DL	Retail products	Price estimation	Future rich features
18	XGBoost / LGBM / RF / MLP	E-commerce	Forecasting/optimization	Ensemble methods
19	Multimodal DRL	Amazon marketplace	Dynamic pricing	Long-term extension
20	AI bibliometric review	Multiple industries	Dynamic pricing review	Current research gaps

Across the literature, three broad directions are visible. First, supervised regression remains a practical formulation for estimating observed prices. Second, ensemble tree models are repeatedly used for structured e-commerce data. Third, advanced dynamic-pricing research increasingly treats pricing as a sequential decision problem using deep reinforcement learning, demand signals and competitor information. The present project is positioned primarily in the first direction while using the second as its main modelling strategy and identifying the third as future work.

13. Research Gaps

13.1 Limited Historical Coverage

The supplied dataset contains only 39 unique dates. This is adequate for a student prototype but insufficient for learning long seasonal patterns or robust annual effects. Future data collection should extend the observation period.

13.2 Uneven Platform Coverage

Platform availability varies substantially. Missing observations may reflect unavailable products, missing collection or platform-specific coverage. A model should distinguish missing price from an actual zero price and should document its imputation strategy.

13.3 Sparse Product-Platform Series

Some product-platform combinations have much shorter histories than others. This creates sparse time series and may limit lag-based features. Research on sparse dynamic pricing highlights cold-start and low-traffic issues as important challenges.

13.4 Lack of Demand and Inventory Variables

The current dataset records prices but not demand, sales volume, stock level, clicks, promotions or customer behaviour. Consequently, the project predicts observed prices rather than estimating an optimal price from demand elasticity.

13.5 Temporal Leakage

Random train-test splitting can allow information from future dates to enter the training set. A time-based split is more appropriate when the intended use is future price prediction. Rolling validation can be added for a stronger evaluation.

13.6 Product Identity and Comparability

Product names are sufficient for the current controlled dataset, but real e-commerce systems need robust product matching. Differences in package size, brand, quantity and product specification can create misleading comparisons if entities are not standardized.

13.7 Interpretability

A practical pricing system should explain which factors contribute to predictions. Feature importance, permutation importance or SHAP can be used after a reliable predictive model is established.

13.8 Real-Time Deployment

A future system would need automated data acquisition, product matching, missing-data handling, timestamp synchronization and model serving. Offline accuracy is only one part of a real-time price-intelligence system.

13.9 Generalization

A model trained on 12 products and a limited time period may not generalize to new products or platforms. Future datasets should contain a larger product catalog and longer time horizon.

14. Proposed ML-Based System

14.1 Architecture

Online Platforms / Historical Data → Data Collection → Cleaning → Long-Format Transformation → Feature Engineering → Regression Model → Predicted Price → Cross-Platform Comparison → Price Insight

14.2 Initial Implementation

1. Load the product-price spreadsheet.
2. Normalize headers and forward-fill dates.
3. Replace “-” and blank price entries with missing values.
4. Convert the wide platform structure to long format.

5. Explore product, platform and temporal price patterns.
6. Create product, platform and time features.
7. Create lag and rolling features using past data only.
8. Use chronological train/validation/test splitting.
9. Train Linear Regression and Decision Tree baselines.
10. Train Random Forest and XGBoost/Gradient Boosting models.
11. Evaluate with MAE, RMSE and R^2 .
12. Inspect residuals and feature importance.
13. Build a simple comparison table showing observed and predicted prices.

14.3 Future Data Acquisition Extension

A later version can automate collection from public APIs or permitted data sources, store timestamped observations and update the model periodically. The system should comply with platform terms and applicable data-use restrictions. Real-time collection should not be treated as a requirement for the initial prototype.

14.4 Scope Justification

The proposed architecture is realistic for a student project because the first stage can be implemented entirely from the provided dataset. It keeps the main contribution on data preparation, machine learning, evaluation and price comparison while leaving live acquisition as future scope.

15. Experimental Design and Evaluation

15.1 Models

Model	Role	Strength	Limitation
Linear Regression	Baseline	Simple and interpretable	Limited nonlinear relationships
Decision Tree Regressor	Baseline	Captures nonlinear rules	Can overfit
Random Forest Regressor	Primary	Robust tabular ensemble	Larger model; tuning required
XGBoost Regressor	Primary	Strong nonlinear learning	Needs tuning and careful validation
SVR	Optional	Effective nonlinear regression for scaling and parameter sensitivity	
LSTM/BiLSTM	Future	Temporal sequence modelling	Needs longer sequences and more data
DRL	Future	Sequential pricing decisions	Requires reward/state design and demand feedback

15.2 Evaluation Metrics

MAE (Mean Absolute Error) measures the average absolute difference between actual and predicted price. **RMSE (Root Mean Squared Error)** penalizes larger errors more strongly. **R²** measures the proportion of variance explained by the model relative to a baseline mean predictor. **RMSLE** can be useful when proportional error matters and price distributions are skewed, provided all target values are non-negative.

15.3 Time-Based Validation

For future price prediction, chronological validation is preferred. Earlier dates should be used for training, a later block for validation and the final block for testing. A rolling-origin evaluation can provide several historical forecasting windows and reduce dependence on a single split.

15.4 Error Analysis

The analysis should examine errors by product and platform rather than only reporting one global metric. Expensive products can dominate RMSE, while very cheap products can be disproportionately affected by percentage error. Reporting both absolute and relative metrics provides a clearer picture.

15.5 Model Selection

The final model should be selected from measured validation performance, stability across time splits, computational cost and interpretability. No model should be assumed to be superior before experimentation. If two models have similar predictive performance, the simpler and more interpretable model can be easier to deploy and explain.

16. Expected Results, Advantages and Limitations

16.1 Expected Results

The expected outcome is a regression model capable of estimating product price from product, platform and historical temporal information. Tree-based ensembles are strong candidates because related e-commerce studies frequently evaluate or report strong performance from Random Forest and boosting models. Actual performance must be measured experimentally on the supplied dataset.

16.2 Advantages

- Data-driven price estimation.
- Cross-platform price comparison.
- Historical price-trend analysis.
- Model comparison using standard regression metrics.
- Feature-importance analysis.

- Potential for future real-time price intelligence.
- Can be implemented with ordinary student computing resources.

16.3 Limitations

- Small dataset and short observation period.
- Uneven platform coverage and missing prices.
- No demand, inventory or customer-behaviour variables.
- Limited product diversity.
- Potential product-matching issues in future external data.
- Results may not generalize to other periods or platforms.
- Dynamic pricing optimization is outside the initial supervised-regression scope.

16.4 Future Scope

- Collect longer historical price series.
- Add discounts, stock, ratings and demand-related features.
- Add automated product matching across platforms.
- Use rolling time-series validation.
- Experiment with LSTM/BiLSTM for longer sequences.
- Investigate multimodal product descriptions and images.
- Explore reinforcement learning for pricing decisions after demand data are available.
- Develop a real-time price comparison dashboard.

17. Conclusion

The reviewed literature demonstrates that machine learning is increasingly used in e-commerce for product-price prediction, price comparison, forecasting and dynamic pricing. Classical regression models provide interpretable baselines, while Random Forest, XGBoost and related ensembles are well suited to structured tabular pricing data. Recent research extends the problem toward time-series deep learning, multimodal price estimation and reinforcement learning for sequential pricing decisions.

The selected product-price dataset provides a practical starting point for a student-level ML project. After cleaning and restructuring, it contains 468 product-date observations and 3,237 observed platform-price records across 12 products, 10 platforms and 39 dates. The major data challenges are uneven platform coverage, missing values and limited historical depth.

The proposed project will focus on supervised regression rather than direct pricing optimization. Linear Regression and Decision Tree will provide baselines, while Random Forest and XGBoost/gradient boosting will be evaluated as primary ensemble candidates. MAE, RMSE and R^2 will be used for evaluation, with time-aware validation to reduce temporal leakage. Feature importance and product/platform error analysis will provide interpretability.

The principal research gaps are limited data coverage, sparse product-platform histories, absence of demand and inventory variables, product matching, temporal leakage and real-time deployment. Addressing these progressively can evolve the project from a historical dataset-based prototype into a broader e-commerce price-intelligence system.

18. References – 20 Research Papers

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Dataset Source

Kaggle dataset: *Comparing Product price on various Online platform*. The local file analyzed in this report is *product prices.xlsx*.

Note: The report uses the supplied spreadsheet as the primary dataset evidence. Literature claims are based on the cited scholarly sources and are intended to support the proposed ML methodology rather than to imply that the cited models have already been tested on this exact dataset.